

What the Largest AI Buyer Says It Bought: An Audit of the 2025 Federal AI Use Case Inventory

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Abstract

Since 2020 every federal agency outside national security has been required to publish an annual inventory of its AI use cases, and since April 2025 the agency itself decides whether each one is “high-impact” and therefore subject to seven minimum risk management practices. The 2025 inventory lists 3,611 use cases from 41 agencies. This paper checks the parts of it that can be checked from the public text.

Three findings are mechanical and exact. Nine percent of the inventory is a name and nothing else: Commerce, TVA and Education published no development stage, no impact determination and no description for 338 rows, and 84 percent of all missing required cells are missing because the agency left the column out of its public file rather than because a cell was blank. Of the 227 deployed use cases agencies themselves marked high-impact, 33 report all six minimum practices complete, 78 report none complete, and 101 report nothing at all because the Department of Veterans Affairs, which holds 94 of them, did not publish those columns. Of 1,480 live use cases, 37 state any numeric performance figure, 9 say what it was measured over, and none states an uncertainty.

The fourth finding needed a blinded reading. Six independent passes rated 844 live use cases against the memo’s own definition and its fifteen presumed-high-impact categories, seeing the description and nothing else. Where the agency said not high-impact, 10 percent read as high-impact, 26 percent after calibrating the passes against a more careful blinded review. Where the agency explicitly overrode a presumption, a blinded reader disagreed with the override 42 percent of the time, and 47 of those 110 overrides share template justification text. Where the agency said high-impact, 19 percent read as not: the Department of Justice lists a CAPTCHA widget and an HR chatbot as high-impact AI, and lists the federal recidivism instrument PATTERN twice, once high-impact and once not. Two preregistered predictions held, three failed and one proved untestable, and one failure reversed the study’s premise: the agencies that self-report zero high-impact use cases are not hiding them, and the agencies that flag the most are also the ones with the most unflagged.

1. Introduction

The United States government is the largest single buyer of AI in the world. Executive Order 13960, signed in December 2020, made it the first to require that every agency outside the national security community publish an inventory of where it uses AI, and the Advancing American AI Act of 2022 wrote the requirement into statute. OMB Memorandum M-24-10 in March 2024 added a governance layer, and OMB Memorandum M-25-21 in April 2025 rescinded and replaced it. The 2025 memo defines “high-impact AI” as AI whose output “serves as a principal basis for decisions or actions with legal, material, binding, or significant effect” on rights, benefits, health, safety, infrastructure or strategic assets. It lists fifteen categories of use that are presumed high-impact, from patient diagnosis to biometric identification to benefits adjudication. It requires seven minimum risk management practices for high-impact AI within 365 days, on pain of the use being discontinued. And it makes the agency the judge of whether its own use case is high-impact, subject only to a requirement that an official document in writing any decision to override a presumption.

The inventory is therefore a self-report at every level. The agency decides what to list, what stage it is at, whether it is high-impact, and whether the practices are done. OMB consolidates what agencies post and publishes the result, with a summary table that treats each agency’s numbers as given. Nobody outside the agency checks whether the description supports the classification, and until now nobody has tried.

This paper does the checking that the public text permits. Some of it is arithmetic: whether the fields the reporting instructions mark as required are filled, what the minimum-practice columns actually say, whether a use case listed in 2024 can be found in 2025, and who the vendor field names. Some of it needs a reader: whether a description of a use case, read by someone who does not know which agency wrote it or what box the agency ticked, falls under the memo’s definition. The design for that part is the one used in two companion audits, of a computer-vision benchmark and of AI claims in corporate annual reports: a rubric written down before any text is read, two independent blinded passes over every sampled row, a calibration pass by a more careful reader who also does not see the verdicts, and a set of mechanical screens that enumerate rather than sample so that the counts they bound are exact.

The results are less tidy than the companion audits, and more interesting for it. Three of the six preregistered predictions failed. The one I expected most confidently, that the agencies reporting zero high-impact use cases would turn out to have the highest rate of unflagged ones, was wrong in the other direction. That is reported in Section 5.3 with the same prominence as the findings that went the way I expected.

2. Related work

The academic literature on federal AI inventories is thin, because the inventories are new and until 2024 were inconsistent enough to defeat analysis. The Government Accountability Office reviewed the first round in 2023 and found that agencies had reported about 1,200 use cases, that the inventories were incomplete and inconsistent across agencies, and that OMB had not yet issued the guidance the executive order required (GAO 2023). Stanford’s RegLab reached a similar conclusion from the 2022

inventories, documenting that a large share of agencies had failed to publish one at all and that the ones published disagreed on what counted as a use case (Lawrence, Cui and Ho 2023). Both studies are about whether the inventories exist and whether they are internally consistent. Neither reads the entries against the governing definition, because in 2022 and 2023 there was no operational definition to read them against.

M-25-21 changed that by supplying one. Its high-impact definition is precise enough that two readers can apply it independently and be scored for agreement, and its fifteen presumption categories are concrete enough to screen for by keyword. That is what makes this study possible and what makes it different from the earlier ones: the question is no longer whether agencies filled in the form but whether what they wrote in the form supports what they ticked.

The method draws on work that is not about government at all. Bean et al. (2025) reviewed 445 language-model benchmarks with 29 expert reviewers and found pervasive failures of construct validity, meaning benchmarks that do not measure what they claim. Raji, Bender, Paullada, Denton and Hanna (2021) argued that the gap between a benchmark’s stated construct and what it actually tests is hidden precisely because nobody writes the construct down. A high-impact determination is a construct-validity judgement of the same kind, made by an interested party, about a system the party built or bought. The audit treats it accordingly.

The two companion studies supply the specific design. The KFuji annotation audit (Shah 2026a) found that a chain of automated passes could produce a confident, internally consistent number that was wrong by nearly a factor of nine, and that only a cheap blinded re-check caught it. The audit of AI claims in annual reports (Shah 2026b) used the same design on text and found the automated passes accurate on the judgement that carried the headline. This study sits between those two outcomes, and Section 6.3 argues that the difference is predictable from how crisp the judgement is.

3. The data

The 2025 inventory is OMB’s consolidation of what each agency posted on its own website, published as a repository with a data dictionary, a validation report and a sourcing summary that records, for each agency, which columns the agency left out of its public file. This study uses the repository at commit 06c7ebe of 14 May 2026: 3,611 individually reported use cases from 41 agencies, plus a separate file of consolidated commercial off-the-shelf uses that is not analysed here. The 2024 consolidated inventory, 2,133 rows at commit 4a29d13, is used only for the year-over-year question. Both files are committed with their hashes.

The inventory is the population of publicly reported individual use cases, not a sample of it. Counts over its structured fields therefore carry no sampling uncertainty. What they carry is whatever the agencies put in the boxes, which is the subject of the study.

Every row has a development stage (Pre-deployment, Pilot, Deployed, Retired, or blank), a high-impact determination (High-impact, Presumed high-impact but determined not, Not high-impact, or blank), a topic area, an AI classification, four free-text fields describing the problem, the benefits, the outputs and the data, a contracting

field with a vendor name, and, for deployed high-impact rows only, nine fields reporting the status of each minimum practice. Of the 3,611 rows, 1,040 are marked Deployed and 440 Pilot. I call those 1,480 rows the live set.

4. Method

4.1 What was fixed in advance

The preregistration in docs/PREREGISTRATION.md was written after the structured fields had been profiled and before any free text was read for rating, any screen was run, any year-over-year matching was attempted, or any vendor string was tabulated. It states which tabulations had already been seen, and it makes no prediction about them. Six predictions were written on the parts not yet examined. The rater instruction was committed to docs/RATER_PROMPT.md before any rater was dispatched, which the companion audit of annual reports did not do and listed as a limitation.

4.2 The rubric

Raters saw seven fields: the use case name, topic area, AI classification, and the four free-text descriptions. They did not see the agency, bureau, development stage, the agency's determination or its justification, the vendor, the PII flag, or any minimum-practice field. Agency and bureau names and their abbreviations, 789 tokens in all, were replaced with [agency].

Part A records two judgements. `category` is the single lettered presumption category from M-25-21 Section 6 that the described use most closely falls into, or none. The fifteen categories were reproduced verbatim in the instruction. `decision_role` is principal if the text describes the AI output as determining or being the main basis for a decision or action about a person, a benefit, an enforcement outcome or physical safety. It is assistive if the output informs, flags, ranks, drafts or recommends for a human who decides, and none if no such decision or action is described. The derived reading, fixed in advance, is `high_impact` when the category is not none and the decision role is not none. The memo states that a high-impact determination "is possible whether there is or is not human oversight," so an assistive role does not exempt. A stricter variant requiring principal is reported alongside as a floor.

Part B records five binary evidence judgements identical to the annual-report audit: `quantified`, `metric_named`, `evaluation_described`, `baseline_given`, `uncertainty_given`. A use case is checkable if the first three are true. The instruction is explicit that a count of users, a budget, a date, a training-set size or a volume of records processed is not a performance figure.

Part C asks whether the text describes the system as `in_use`, `not_yet`, or `unclear`, to be compared afterwards with the agency's own stage.

4.3 The sample

The rated sample is drawn from the 1,450 live rows with a non-empty description: every row the agency marked High-impact (250), every row marked Presumed but not (71), every live row with a blank determination (30), and up to 30 rows per agency

from those marked Not high-impact under seed 20260926 (493). That is 844 rows, each assigned to two of six raters by a fixed rule, 1,688 judgements, 280 to 282 per rater.

4.4 The passes and the calibration

The six passes ran as independent Claude Sonnet subagents with no shared context. A schema validator confirmed every line, every assigned row, no duplicates and no omissions before anything else was computed.

Calibration has three parts. The first re-reads a seeded sample of 110 rated rows, 40 that both passes read as high-impact, 40 that both read as not, and 30 where they disagreed, presented in a shuffled order with eight rows repeated under a second identifier as self-consistency controls. I read all 118 displays and recorded a category, role and stage for each before the key was opened. The second part is the numeral screen: a performance figure needs a number, so every live row was screened for one and every candidate read blind. The third is the presumption keyword screen, a fixed regular expression per category built from the memo's own words and frozen before it was run, whose precision the blinded readings then measure.

4.5 Statistics

Population counts carry no interval. Rates from the rated sample carry Wilson intervals, and where rows nest within agencies, a percentile bootstrap resampling agencies with 10,000 replicates. The calibrated rate for the not-high-impact stratum is bootstrapped jointly over the calibration units. Agreement is observed agreement and Krippendorff's alpha for nominal data. `src/stats.py` runs its own checks and `src/verify_paper.py` recomputes every number in this paper from the raw data.

5. Results

5.1 A tenth of the inventory is a name and nothing else

Against the fields the reporting instructions mark as required for each row's stage, the inventory is 91.3 percent complete: 38,031 of 41,642 required cells are filled. The number that matters is what the other 3,611 cells are. OMB's own sourcing summary records which columns each agency dropped from its public file, and 3,033 of the missing cells, 84 percent, are in columns the agency did not publish at all. Only 578 are cells left blank inside a column the agency did publish. The public inventory's incompleteness is mostly a decision not to publish, not a failure to answer.

The extreme case is 338 rows, 9.4 percent of the inventory, with no development stage and no impact determination. All of them come from three agencies. The Department of Commerce published 223 rows containing a name, a bureau and a contact address, and for 25 of them a one-line problem statement. TVA published 59 rows with a name, a topic and a benefits line. Education published 56 with a name, a problem and a benefit. OMB's summary table nevertheless reports Commerce as having 223 use cases, 223 of them deployed and none high-impact, which is not supported by anything in the file. Thirty of the 41 agencies omitted at least one required column.

5.2 The passes agree, and the disagreement is where the definition is soft

Across the 844 rated rows the two passes gave the same category 91.7 percent of the time (alpha 0.850), the same decision role 88.2 percent (alpha 0.777), and the same derived high-impact reading 93.8 percent (alpha 0.864). The stricter principal-only reading agreed 99.2 percent, which mostly reflects how rarely either pass used principal: 57 of 1,688 judgements.

The 52 disputed readings sit where the rubric asks a graded question. Twenty-three of them are one pass calling a law-enforcement category and the other calling none, on sentences about database tools, entity resolution and investigative search where the text does not say whether anything downstream depends on the output. The blinded reviewer resolved 23 of the 30 disputed rows in the calibration sample as high-impact.

Part B agreed as it did in the annual-report audit: quantified 99.9 percent (alpha 0.975), the other four between 96 and 99 percent. Part C did not agree. The stage reading matched 68.1 percent of the time with alpha 0.372, and Section 5.6 reports why.

5.3 The blinded reading against the agency's box

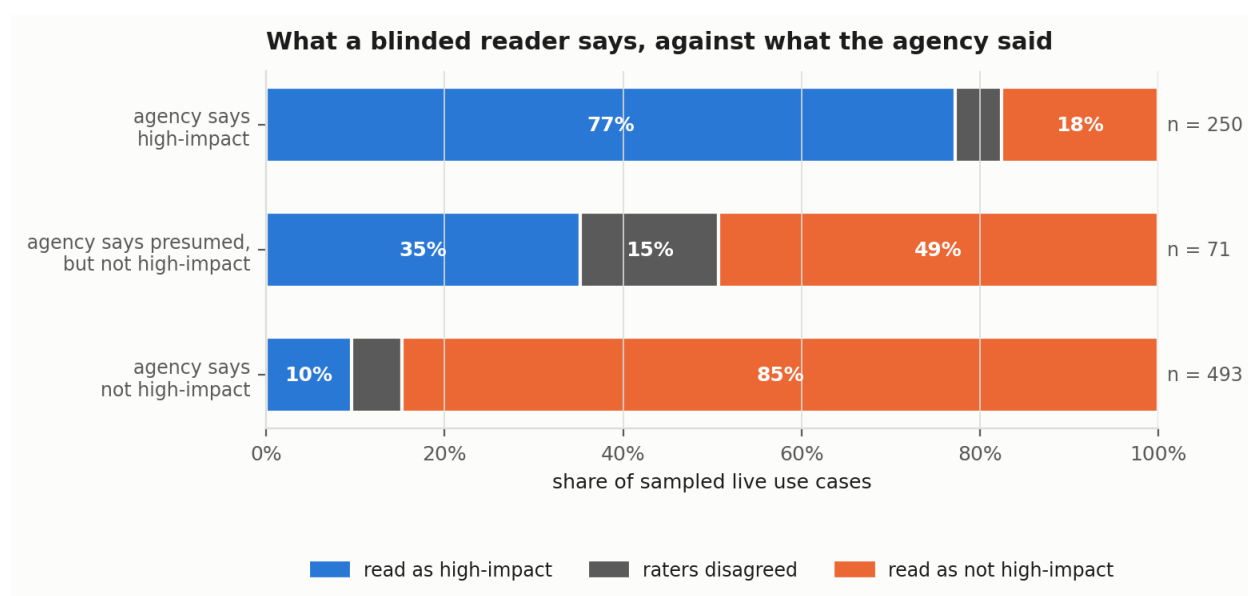


Figure 1: What a blinded reader says about a use case, against what the agency said. Sampled live rows, grouped by the agency's own determination. Blue is the share both passes read as high-impact under the derived rule, orange the share both read as not, grey the share on which the two passes disagreed.

Of the 250 sampled rows agencies marked High-impact, both passes read 193 as high-impact and 44 as not, with 13 disputed: 81.4 percent agreement with the agency (Wilson 76.0 to 85.9). Of the 493 rows marked Not high-impact, both passes read 47 as high-impact, 418 as not, 28 disputed: 10.1 percent of the agreed rows read as high-impact (Wilson 7.7 to 13.2, agency bootstrap 6.6 to 14.0). Of the 71 rows where

the agency invoked the override, “presumed high-impact but determined not,” both passes read 25 as high-impact and 35 as not, 11 disputed: 41.7 percent (Wilson 30.1 to 54.3).

The calibration moves the second of those numbers a long way. On the 40 calibration rows both passes had read as not high-impact, I read 6 as high-impact, a survival rate of 85 percent, and I resolved 23 of 30 disputed rows as high-impact. Applying those rates, the calibrated share of not-high-impact rows that read as high-impact is 26.4 percent, with a calibration bootstrap interval from 17.7 to 36.4. The corresponding survival on the high side was 39 of 40, so the calibrated agreement with an agency’s high-impact determination stays at 81.9 percent, and the calibrated disagreement with an override rises to 53.6 percent.

I report the raw and calibrated figures side by side rather than choosing, because the gap between them is itself a result about the method (Section 6.3). The conservative reading is that one in ten use cases an agency says is not high-impact reads as high-impact to two independent readers who agree with each other. The calibrated reading is one in four. Either way, extrapolating the per-agency sampled rates to the 1,129 live not-high-impact rows in the population gives an estimated 106 use cases, against the 250 live use cases agencies themselves marked high-impact, on the raw rate alone.

Two preregistered predictions were about this. The first, that at least 15 percent of not-high-impact rows would read as high-impact, fails on the raw rate and holds on the calibrated one. The preregistration did not say which, and I score it as not cleanly held. The second was that disagreement would concentrate in the agencies that self-report zero high-impact use cases, on the theory that an agency reporting none among hundreds of live systems is under-classifying. It failed, and in the opposite direction.

The three large agencies reporting zero high-impact live use cases, HHS, Interior and Treasury, had 6 of 88 sampled not-high-impact rows read as high-impact, 6.8 percent. The three that report the most, VA, Justice and Homeland Security, had 16 of 80, or 20.0 percent. The bootstrap difference is 13 points the wrong way, with an interval from 4 to 23 that excludes zero. VA, which marks 64 percent of its live use cases high-impact, had 8 of 27 of its remaining rows read as high-impact. SSA, which marks 29 percent, had 6 of 20. HHS, with 255 live use cases and no high-impact ones, had 1 of 28.

The explanation is in what the agencies do. HHS’s inventory is dominated by research models and internal tools at NIH, CDC and FDA. Interior’s is dominated by USGS science. Treasury’s is IRS document processing. Those are the agencies where a blinded reader, handed the description, finds nothing that touches a person’s rights or benefits. VA’s inventory is medical devices and clinical decision support, Justice’s is investigative tools, Homeland Security’s is border and immigration systems. Those agencies classified a lot as high-impact because a lot of what they do is, and what they left unflagged is drawn from the same distribution. The prediction assumed that agency-level variance in self-reporting was a measurement artefact. It is mostly signal, and the residual disagreement scales with the agency’s exposure rather than with its reticence.

The categories the readers assigned to the rows they read as high-impact tell the same story: healthcare (f) 97 times, law enforcement (j) 89, benefits and services (m)

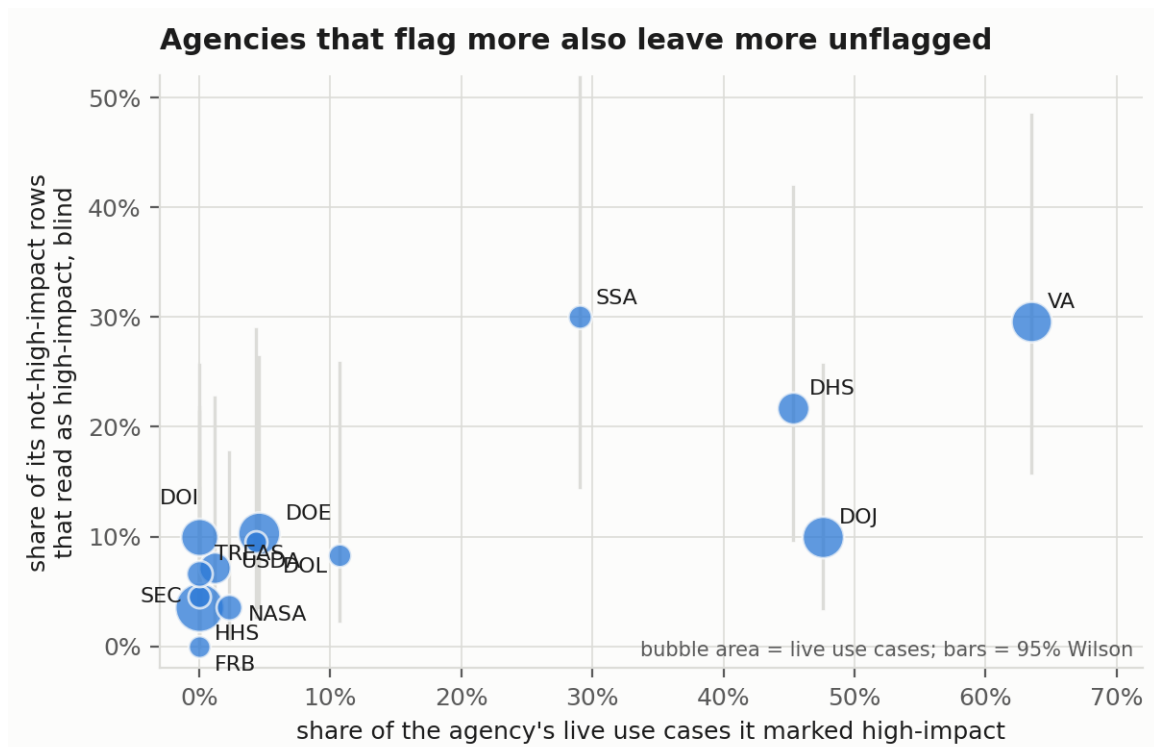


Figure 2: Per agency, the share of its not-high-impact rows that read as high-impact under blinded review, against the share of its live use cases it marked high-impact itself. Agencies with at least 20 sampled rows. The unlabelled point at 10 percent is State.

38, safety-critical infrastructure (a) 17. Everything else is single digits.

Three individual cases are worth naming, because every judgement in this study is published and a reader can check them. Justice lists the Prisoner Assessment Tool Targeting Estimated Risk and Needs, PATTERN, the federal Bureau of Prisons' recidivism instrument, in two rows. One says the tool "uses pre-defined rules to score an inmate's recidivism risk level" and is marked Not high-impact. The other says the intended use is "to predict the risk of recidivism for incarcerated adults" and is marked High-impact. Both passes read both rows as category j, assistive, high-impact. M-25-21's presumption list names "determinations related to recidivism" in so many words. In the other direction, Justice marks as high-impact a row named Cloudflare Turnstile, which is a CAPTCHA replacement, a row named Building Automation Systems, and a Chatbot to Answer Internal Employee Policy Queries. Both passes read all three as no category. Thirty-two of Justice's 314 rows invoke the override, and all 32 carry the same justification sentence, verbatim.

That last point generalises. Of the 110 rows across the inventory where an agency overrode a presumption, 47 share a justification text used three or more times, and 32 of those are Justice's single sentence. The memo requires "written documentation" for an override. The inventory suggests that at some agencies the documentation is a template.

5.4 Thirty-seven use cases give a number, nine say what it means, none gives an interval

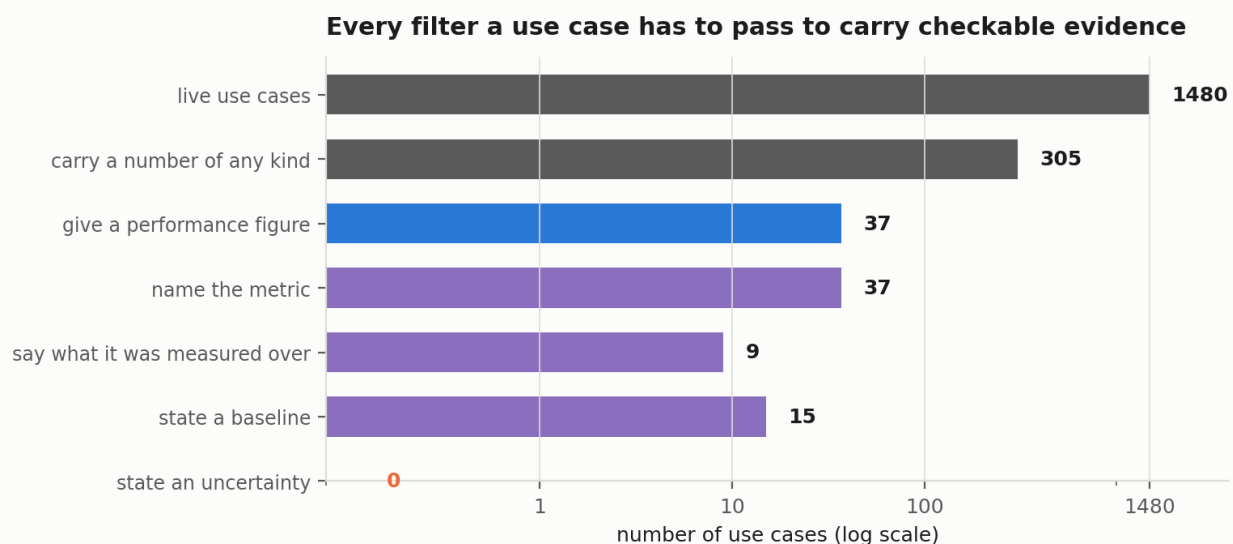


Figure 3: Every filter a live use case has to pass before it carries evidence an outsider could check. Log scale. Zero is shown as a number with no bar.

The numeral screen, in its extended form (Amendment 1), flagged 305 of the 1,480 live rows as carrying a number of any kind. I read all 305 blind. Thirty-seven contain a numeric performance figure attributed to the AI, 2.5 percent of live use cases (Wilson 1.8 to 3.4). All 37 name the metric, 15 give a baseline, 9 say what data, population or period the figure was measured over, and none states an interval, a sample size or an error estimate. Nine use cases in the federal inventory, 0.61 percent (Wilson 0.32 to 1.15), are checkable in the study's sense.

The two automated passes, on the 185 candidates that fell inside the rated sample, marked 20 rows quantified and I marked 19. The 19 coincide, and the one the passes added was a NASA toolkit claiming “orders-of-magnitude reductions” without a number. Four rows the passes marked quantified were outside the original screen, and they are why the screen was extended.

The nine checkable claims are what good disclosure looks like at the scale a form allows. VA reports that randomised implementation of computer-aided detection during colonoscopy across its facilities produced “a statistically significant 21% increase in the odds of adenoma detection” against colonoscopy without it. Interior reports a convolutional network reaching 78 percent validation accuracy on stream PFAS data against 65 percent for logistic regression and boosting. VA reports its Privacy Act automation cut response time on lower-complexity cases “from over a month to less than 4 days” across more than 25,000 cases since deployment. These are not audited numbers. They are numbers with enough context that an auditor would know where to start, and there are nine of them.

Most of the 37 are not like that. “Reduces time spent searching course materials by over 80%.” “Up to 60-80% reduction in inquiry volume.” “Detects cooling towers

approximately 600 times faster than manual searches.” A figure, a direction, and nothing to measure it against. And many of the rows I did not count are ones that state a volume: 2 million actions in a year, 25,000 cases processed, 15,000 emails handled. A volume tells you the system is busy. It does not tell you whether it is right.

5.5 Of 227 deployed high-impact use cases, 33 report all six practices done

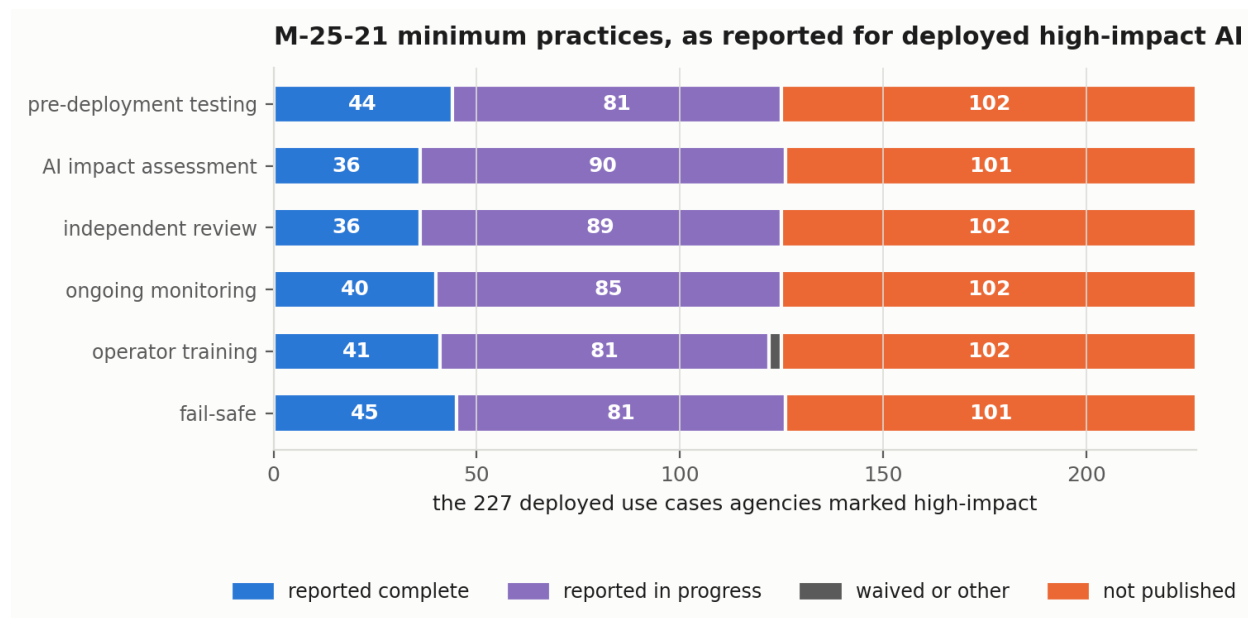


Figure 4: The status of each M-25-21 minimum practice, as reported, for the 227 deployed use cases their agencies marked high-impact. Orange is rows whose agency did not publish the column at all.

M-25-21 gave agencies 365 days from 3 April 2025 to document six practices for every high-impact use: pre-deployment testing, an AI impact assessment, an independent review, ongoing monitoring, operator training and a fail-safe. Inventories were due to OMB on 22 December 2025, three and a half months before that deadline, so in-progress answers are not violations. The question is what agencies said.

For the 227 deployed use cases marked high-impact, the six practices are reported complete on between 36 and 45 rows each. In-progress is reported on 81 to 90. Between 101 and 102 rows report nothing, and that number is not a mystery: VA holds 94 of the 227 and did not publish any of the minimum-practice columns, and NCUA, FDIC and NASA account for the other 7. Thirty-three use cases report all six complete, 26 of them at Homeland Security and 7 at SSA. Seventy-eight report every practice as in-progress and none complete, and 73 of those are Justice’s entire deployed high-impact inventory.

Read by agency, the picture is three different postures. Homeland Security reports the practices done on 26 of 38. Justice reports all 73 of its in progress. VA, with the largest high-impact deployment in government, reports nothing in public. OMB’s sourcing summary confirms that VA’s file simply lacks the columns, so this is non-publication rather than non-compliance, but the memo’s own text says the inventory

is one of the places agencies must be “prepared to report” implementation, and the public cannot tell from VA’s inventory whether a single one of its 94 high-impact clinical and benefits systems has been tested.

5.6 Two in five 2024 entries cannot be found by name, and the description cannot tell you what is deployed

The 2025 instructions require a use case that was retired to be reported as retired in the following year’s inventory before it can be dropped. The 2024 file carries no identifier, so matching is by agency and normalised name. Of the 2,133 rows in 2024, 171 belong to nine agencies absent from the 2025 consolidation, including OPM, USAID and CFPB. Of the 1,962 that could be matched, 1,195 have an exact name match in 2025 in any stage, and 767, 39.1 percent, do not. That satisfies the preregistered prediction as written, which was that at least 20 percent would have no name match. It is also an overstatement of what disappeared.

I reviewed a seeded sample of 60 unmatched rows against their agency’s 2025 names. Twenty-seven are renames or restructurings, some explicit (HUD’s “Automated Draft Narrative Reports, previously Automating Draft Counterparty Credit Narrative Reports”) and some obvious (Education’s five identical 2024 rows named “Generative AI Usage” became a dozen rows named by function). Five were already marked retired in 2024 and had no obligation to reappear. Nine are commercial products such as Camtasia, Adobe and CrowdStrike that the 2025 rules moved to a separate consolidated file. Nineteen have no plausible successor. Scaling, the share of matchable 2024 rows that disappeared without a retirement notice is at most 12 percent excluding the commercial products and 18 percent including them, and the rename rate is a lower bound because one rename was found outside the similarity shortlist (Amendment 4).

On deployment itself, the structured field says 28.8 percent of the inventory is deployed and 41.0 percent is live. Part C was meant to test whether the descriptions support the stage field, and it failed as an instrument. The passes agreed on the stage reading with alpha 0.37, and where a pass said `in_use` the blinded reviewer agreed on 33 of 150 judgements, reading 114 as unclear. The instruction said that present-tense capability with no operational indicator is unclear, and the passes read present-tense capability as `in_use` anyway. I do not think the passes were careless. I think the descriptions are written in a register that does not distinguish “this system does X” from “this system is doing X for us now,” and a rater has to choose. The preregistered prediction is recorded as untestable. What survives is the negative finding: the free text of the inventory does not let a reader verify the stage field.

5.7 Microsoft, then everyone else

Of the 1,480 live use cases, 552 are marked as purchased from a vendor, 387 as built with both contracting and in-house resources, 505 as built in-house, and 36 leave the field blank. Of the 939 with vendor involvement, 625 name one. The 378 distinct strings normalise to 375 parents, most of them appearing once.

Microsoft is named on 110 of the 625, 17.6 percent, mostly through Azure OpenAI

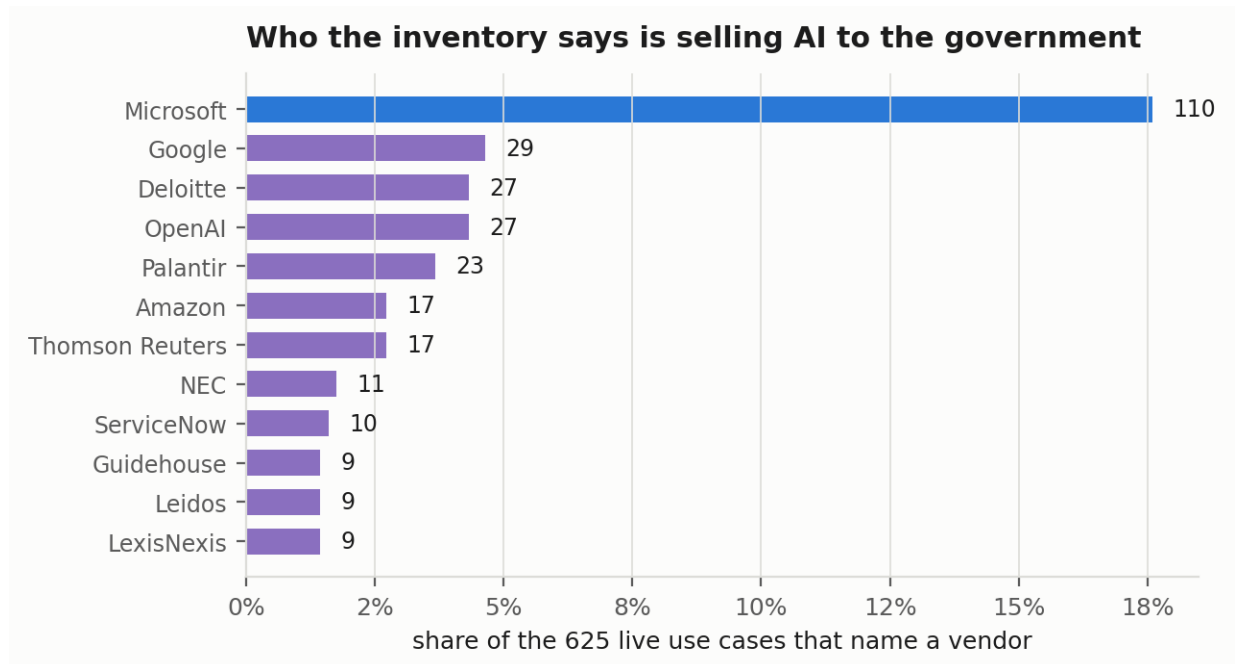


Figure 5: The twelve vendors named most often on live use cases, after normalising 378 distinct vendor strings. Share of the 625 live use cases that name at least one vendor.

and Copilot. Google follows with 29, Deloitte and OpenAI with 27 each, Palantir with 23, Amazon and Thomson Reuters with 17. The top five vendors hold 34.6 percent of the named rows and the top ten 44.8 percent. The preregistered prediction was that the top five would hold more than half. It failed. The vendor field describes a market that is concentrated at the top and long-tailed underneath, and the concentration is uneven across agencies: Microsoft is named on 50 of Energy’s 127 vendor-named rows and 8 of HHS’s 137.

Two caveats limit this section. VA did not publish its vendor column at all, so the agency with the most deployed high-impact AI contributes nothing here, and “Microsoft” on a row that says “Azure OpenAI” is a channel rather than a model developer. What the inventory can support is the narrower claim that when a federal agency names who it bought AI from, one name in six is Microsoft.

6. What this means

6.1 The inventory is a disclosure instrument that is mostly working and partly not

It is easy to read an audit like this as an indictment, and it is not one. The 2025 inventory is a large improvement on 2024: the fields are standardised, OMB published the validation work, the free text is long enough to read, and for 91 percent of required cells someone filled in an answer. Ninety-three percent of the time two independent readers agree on whether a description falls under the memo’s definition, and 81 percent of the time they agree with the agency that marked it high-impact.

Self-classification under M-25-21 is not random.

What the audit finds is three specific places where the instrument gives way. The first is non-publication: 84 percent of what is missing is missing because an agency dropped a column, and the agencies that dropped the most are the ones whose columns matter most, VA on minimum practices and Commerce on everything. The second is the override. Where an agency has said “this is presumed high-impact but I have determined it is not,” a blinded reader disagrees 42 to 54 percent of the time, and nearly half of the written justifications the memo requires are template text. The third is the soft edge of the definition itself, where “assistive” tools in law enforcement and benefits contexts are classified one way at one agency and another way at the next, or both ways at the same agency for the same tool.

6.2 For the buyer

The business reading of this inventory is not the vendor table. It is Section 5.4. The federal government, buying from 375 named vendors, was able to state a performance figure for 37 of 1,480 live systems and a checkable one for 9. That is a slightly lower quantified rate than the corporate annual reports audited in the companion study (2.7 percent of capability claims) and a higher checkable rate (0.6 percent of live use cases against 0.17 percent of AI sentences), and the comparison is loose because the units differ. What the two audits agree on is the shape: the number is almost never there, and when it is there, the context needed to interpret it is almost never with it. A buyer with the government’s leverage produced the same disclosure gap as the sellers. The gap is not a property of sellers.

6.3 The calibration moved this time, and the reason is predictable

In the annual-report audit the calibration pass found the automated passes essentially exact on the headline judgement. Here it moved the headline from 10 percent to 26. The difference is in the judgement. Whether a sentence contains a numeric performance figure is a question two readers settle by pointing at the characters, and on that question the passes here were as exact as before: 19 of my 19 quantified calls in the sample matched theirs. Whether a description of an investigative database “describes a decision or action about a person” is a graded question with a soft threshold, and on graded questions a model asked for a binary leans one way and keeps leaning. The passes leaned toward none. I leaned toward reading a law-enforcement search tool as feeding an enforcement decision, because that is what such tools are for. Neither reading is wrong on its face, and the honest report is both numbers with the survival rates that connect them.

The rule this suggests is the one the companion audit proposed: an automated pass can be trusted in proportion to how crisply its question can be settled by pointing at the input. On the crisp questions, the number, the baseline, the interval, this study did not need a calibration pass. On the graded one it did, and the correction was a factor of 2.6. A study that ran only the automated passes on the graded question would have reported the smaller number with a tight interval and been wrong about the width.

6.4 What would fix it

Three changes to the inventory would remove most of what this audit found, and none requires new authority.

Publish the columns. OMB already records which columns each agency omitted. It could decline to count an agency's rows as reported until the required columns are present, which would end the situation where the summary table credits Commerce with 223 deployed use cases on the strength of 223 names.

Require the override justification to be specific. A sentence that says the output "does not serve as a principal basis for decisions" restates the definition. It does not apply it. Requiring the justification to name who receives the output and what they do with it would make the 110 overrides reviewable, and would have caught the two PATTERN rows.

Ask for the number with its context. The form already asks agencies to describe the data used to evaluate the model. It could ask, for deployed use cases, for one performance figure, the metric, the population it was measured on, and the comparison. Nine agencies managed that unprompted. The others were not asked.

7. Limitations

The blinded reading is of the description, not the system. A use case whose description understates what it does will read as lower-impact than it is, and one whose description is aspirational will read higher. Section 5.6 shows the descriptions cannot even reliably convey whether the system is running. The reading measures the consistency between what the agency wrote and what it ticked, and nothing about the system itself.

The calibration units are few. The 26 percent calibrated rate rests on 6 of 40 "not" calibration rows flipping and 23 of 30 disputed rows resolving high, and the bootstrap interval is 18 to 36 for that reason. The raw rate is reported beside it.

The presumption categories overlap and the rubric forced one. A visa-fraud facial recognition system is category j, k or l depending on emphasis. The derived reading does not depend on which, but the category counts in Section 5.3 do.

The redaction was imperfect. Product names, bureau acronyms not in the redaction list, and phrases like "Veterans" identify the agency in some rows. The 2025 inventory is public and the agency field is one join away, so the blind here protects the reading from the agency's determination, not from the agency's identity.

The 2024 match is by name. The rename review used a similarity shortlist and found one rename outside it, so the disappearance figure is an upper bound with a known direction of error.

The raters are models. Six passes of Claude Sonnet and a calibration pass by Claude Opus 5, the model orchestrating the study. No human read these rows except me through that model, and I have a stake in the result, which is why the duplicate controls, the key files and every judgement are published. A replication with human

coders would be cheap and would settle whether the graded judgement in Section 6.3 is one humans agree on either.

8. Conclusion

The 2025 Federal AI Use Case Inventory lists 3,611 systems. For 338 of them the government published a name. For the 227 it calls deployed and high-impact, it published a complete set of minimum-practice answers for 33 and no answers at all for 101. For the 1,480 it calls live, it gave a performance number for 37 and a checkable one for 9. And when two readers applied the government’s own definition of high-impact to the government’s own descriptions without knowing what the government had decided, they disagreed with a “not high-impact” determination between one time in ten and one time in four, disagreed with an explicit override roughly half the time, and found the same recidivism tool classified both ways in the same agency’s file.

The premise I brought to the study, that the agencies reporting zero high-impact AI were hiding it, was wrong. They report zero because their AI is science and paper-work. The unflagged high-impact systems are at the agencies that already flag the most, because that is where the high-impact systems are. That is a better finding than the one I predicted, because it points at the definition’s soft edge rather than at any agency’s motives, and a soft edge can be sharpened.

Data and code

Everything is at <https://github.com/ishaanshah101/fed-ai-inventory-audit>.

`src/frame.py` loads the inventory and records provenance, `src/build_rating_sets.py` samples, redacts and assigns, `src/validate_ratings.py` checks the rater output, `src/agreement.py` computes agreement, `src/build_calibration.py` builds the blinded calibration materials, `src/numeral_screen.py` and `src/presumption_screen.py` are the two mechanical screens, `src/calibrate.py` unblinds, `src/completeness.py`, `src/churn.py` and `src/vendors.py` cover RQ1, RQ5 and RQ6, `src/analyze.py` produces every number here, `src/figures.py` draws the figures, and `src/verify_paper.py` recomputes every number quoted in this paper and the README from the raw data and fails on any mismatch. The rater instruction is `docs/RATER_PROMPT.md`, the preregistration with its five amendments is `docs/PREREGISTRATION.md`, and the raw inventory files, the blinded item files, all 1,688 judgements, the calibration materials with their keys and my own judgements are under `data/`.

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